

Primordial Black Hole Evaporation Threshold from the Projective Dynamic Logo Framework: A Falsifiable Prediction for Fermi-LAT

Cédric Laubscher

Independent researcher, Switzerland

cedriclaubscher.ch | zenodo.org/communities/pdl-framework

26 April 2026

Document D45 of the PDL corpus | CC BY 4.0

Abstract

The Projective Dynamic Logo (PDL) framework derives all fundamental physical constants from four axioms on finite signed graphs, without presupposing spacetime, particles, or fields [1, 2, 3]. A central result of the framework is that the effective gravitational coupling is density-dependent: $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$, where $\sigma(N) = 1 - (1 - \kappa_{\text{PDL}})^N$ and $\kappa_{\text{PDL}} = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ is an unconditional theorem of the axioms [4, 5]. At the epoch of primordial black hole (PBH) formation ($N = N_{\text{CMB}} = 40$, derived from the neutron quintuplet without cosmological input [6]), this coupling suppression extends PBH lifetimes by a factor $\sigma(40)^{-2} = 1.4007$. The resulting shift in the PBH survival mass threshold is $M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* = 1.1189 M_{\text{GR}}^*$, corresponding to an upward shift of +11.89% relative to the standard general-relativistic value $M_{\text{GR}}^* \approx 5.1 \times 10^{14} \text{ g}$. This prediction implies an excess of Hawking radiation flux in the isotropic gamma-ray background (IGRB) from PBHs in the mass window $[5.10, 5.71] \times 10^{14} \text{ g}$ that are still active according to PDL but are predicted to have already evaporated by standard general relativity. A complete calculation of the multi-component Hawking spectrum (direct emission, secondary radiation, final-state radiation, and in-flight positron annihilation), performed with the public pipeline GAMMAPBHLOTTER [7] built on BLACKHAWK [8] spectral tables, locates the spectral peak at $E_{\text{peak}}(M_{\text{GR}}^*) \approx 130.1 \text{ MeV}$ under standard GR and at $E_{\text{peak}}(M_{\text{PDL}}^*) \approx 117.5 \text{ MeV}$ under PDL, a fractional displacement of -9.65% , superseding the cruder primary-emission-only estimate ($\approx 90\text{--}105 \text{ MeV}$) used in earlier drafts of this prediction. The prediction is fully determined by a single unconditional theorem (κ_{PDL}) and contains no free parameter beyond the standard-model degrees of freedom entering the Hawking spectrum. We identify the precise observational test using the public codes BLACKHAWK [8], GAMMAPBHLOTTER [7] and ISATIS [9] together with the Fermi-LAT IGRB measurements of Ackermann et al. [10] and the constraint compilation of Carr et al. [11], and invite the community to carry it out.

We note that a very recent independent analysis of the same IGRB data [12] constrains the PBH abundance near this mass to $f_{\text{PBH}} \lesssim 10^{-10}$, which may render the present signal too faint to discriminate regardless of which cutoff is correct; this is discussed in Section 7.

Item	Status
$\kappa_{\text{PDL}} = 310\varphi/11017$	Unconditional theorem of C1–C4 [13, 5]
$N_{\text{CMB}} = 40$	Theorem (neutron quintuplet, no cosmological input) [6]
$\sigma(40) = 1 - (1 - \kappa_{\text{PDL}})^{40}$	Exact closed-form expression
$M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^*$	Unconditional corollary [14]
Numerical value $M_{\text{PDL}}^* = 5.7063 \times 10^{14} \text{ g}$	Computed (this document, verified)
Shift +11.89%	Computed (this document, verified)
Excess flux, peak at 130.1 (GR) vs 117.5 MeV (PDL)	Computed (full BLACKHAWK/GAMMAPBHPlotTER)
Test via BLACKHAWK/GAMMAPBHPlotTER/ISATIS	Spectral computation performed (this document); co

Table 1: Epistemic status of the main claims in this document.

Contents

1	Introduction	4
2	The PDL effective gravitational coupling at the PBH epoch	4
2.1	The coupling suppression factor	4
2.2	Impact on the Hawking evaporation lifetime	5
3	The PDL survival threshold	5
3.1	Derivation	5
3.2	Numerical verification	6
4	The observational signature	6
4.1	Physical picture	6
4.2	Spectral localisation	7
4.3	Nature of the signal	8
4.4	Additional prediction: entropy suppression	8
5	How to test this prediction	8
6	Relation to the PDL causal chain	9
7	Discussion	10
8	Conclusion	11

1 Introduction

The Projective Dynamic Logo (PDL) framework [1, 15] derives fundamental physical constants from four axioms on finite signed graphs without presupposing any background spacetime, particles, or fields. The minimal stationary closure under these axioms is the complete graph K_4 on four vertices and six edges [16], identified with the electron prototype. The proton is the minimal hierarchical composite, characterised by the unique integer quintuplet $(n_u, n_d, r_{\text{val}}, R_{\text{sea}}, R_{\text{tot}}) = (24, 28, 930, 10087, 11017)$ [17, 18].

A central structural result of the framework is the density-dependent effective gravitational coupling [19, 20, 4]:

$$G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}, \quad \sigma(N) \equiv 1 - (1 - \kappa_{\text{PDL}})^N, \quad (1)$$

where N is the number of coherent closure units in the local environment, and $\kappa_{\text{PDL}} = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ is an unconditional theorem of the PDL axioms [13, 5]. Newton's gravitational constant G_{Newton} is recovered as the $N \rightarrow \infty$ limit $G_{\text{Newton}} = G_{\text{PDL}}$, whilst at any finite density the coupling is suppressed by $\sigma(N) < 1$.

This density dependence has direct consequences for primordial black holes (PBHs), which formed at early cosmological epochs when the local closure density was characterised by $N = N_{\text{CMB}} = 40$ [6, 21]. Because the Hawking evaporation lifetime scales as $\tau_{\text{BH}} \propto G^{-2}$, the suppression of the gravitational coupling at that epoch implies longer PBH lifetimes and hence a higher survival mass threshold than predicted by standard general relativity (GR).

This document derives the PDL prediction for the PBH survival threshold M_{PDL}^* , characterises the resulting observational signal in the isotropic gamma-ray background (IGRB), and identifies the precise test that the community can perform with existing Fermi-LAT data and public spectral codes.

2 The PDL effective gravitational coupling at the PBH epoch

2.1 The coupling suppression factor

The cosmological epoch relevant to PBH formation and early evolution is characterised in PDL by the closure density $N_{\text{CMB}} = 40$, derived from the neutron quintuplet structure alone [6]:

$$N_{\text{CMB}} = \lfloor \Gamma_n \rfloor = \lfloor 40.102 \rfloor = 40, \quad \Gamma_n = 6\mu_n - R_{\text{tot}}(n), \quad (2)$$

where μ_n is the neutron's coherence cycle count and $R_{\text{tot}}(n) = 10992$ is its total relational budget. No cosmological parameter is used as input.

The PDL surface fraction is $\kappa_{\text{PDL}} = R_{\text{surf}}/R_{\text{tot}} = 310\varphi/11017$, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio. This is an unconditional theorem of axioms C1–C4 [5]. Numerically:

$$\kappa_{\text{PDL}} = \frac{310\varphi}{11017} \approx 0.045529. \quad (3)$$

The suppression factor at the CMB epoch is therefore:

$$\sigma(40) = 1 - (1 - \kappa_{\text{PDL}})^{40} \approx 0.844935, \quad (4)$$

and the effective gravitational coupling at that epoch is:

$$G_{\text{eff}}(N_{\text{CMB}}) = \sigma(40) \cdot G_{\text{PDL}} \approx 0.8449 G_{\text{Newton}}. \quad (5)$$

2.2 Impact on the Hawking evaporation lifetime

The Hawking evaporation lifetime of a Schwarzschild black hole of mass M is [22, 23]:

$$\tau_{\text{BH}}(M) = \frac{5120 \pi G^2 M^3}{\hbar c^4 f(g^*)}, \quad (6)$$

where $f(g^*) \approx 3.8$ for the Standard Model degrees of freedom $g^* = 106.75$ [11]. Because $\tau_{\text{BH}} \propto G^2$, replacing G by $G_{\text{eff}}(N_{\text{CMB}}) = \sigma(40) \cdot G$ yields:

$$\tau_{\text{PDL}}(M) = \frac{\tau_{\text{GR}}(M)}{\sigma(40)^2}, \quad (7)$$

so PDL lifetimes at the PBH epoch are longer than GR lifetimes by a factor:

$$\frac{\tau_{\text{PDL}}}{\tau_{\text{GR}}} = \sigma(40)^{-2} = 1.4007. \quad (8)$$

3 The PDL survival threshold

3.1 Derivation

The standard GR survival threshold M_{GR}^* is defined by $\tau_{\text{GR}}(M_{\text{GR}}^*) = t_{\text{U}}$, where $t_{\text{U}} \approx 13.8$ Gyr is the age of the universe [24]. The canonical value is [23, 11]:

$$M_{\text{GR}}^* \approx 5.1 \times 10^{14} \text{ g}. \quad (9)$$

The PDL threshold M_{PDL}^* is defined by $\tau_{\text{PDL}}(M_{\text{PDL}}^*) = t_{\text{U}}$. Using equation (7):

$$\frac{\tau_{\text{GR}}(M_{\text{PDL}}^*)}{\sigma(40)^2} = t_{\text{U}} \quad \Rightarrow \quad \tau_{\text{GR}}(M_{\text{PDL}}^*) = \sigma(40)^2 t_{\text{U}}. \quad (10)$$

Since $\tau_{\text{GR}} \propto M^3$, this gives:

$$\frac{M_{\text{PDL}}^*}{M_{\text{GR}}^*} = \left(\frac{t_{\text{U}}}{\sigma(40)^2 t_{\text{U}}} \right)^{1/3} = \sigma(40)^{-2/3}. \quad (11)$$

Falsifiable prediction — PDL survival threshold

$$M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* = 1.11888 M_{\text{GR}}^* \approx 5.706 \times 10^{14} \text{ g}. \quad (12)$$

The shift relative to the GR value is +11.89%, derived from a single unconditional theorem (κ_{PDL}) with no free parameter.

3.2 Numerical verification

All numerical values are derived from the exact expression $\kappa_{\text{PDL}} = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$:

Quantity	Value	Source
φ	$(1 + \sqrt{5})/2$	exact
κ_{PDL}	$310\varphi/11017 \approx 0.04552878$	[5]
$\sigma(40)$	$1 - (1 - \kappa_{\text{PDL}})^{40} \approx 0.84493508$	equation (4)
$\sigma(40)^{-2}$	1.400726	
$\sigma(40)^{-2/3}$	1.118882	equation (11)
M_{GR}^*	$5.100 \times 10^{14} \text{ g}$	[23, 11]
M_{PDL}^*	$5.706 \times 10^{14} \text{ g}$	equation (12)
$\Delta M^*/M_{\text{GR}}^*$	+11.89%	

Table 2: Numerical values entering the PDL threshold prediction. All quantities flow from the single unconditional theorem $\kappa_{\text{PDL}} = 310\varphi/11017$ and the derived value $N_{\text{CMB}} = 40$.

Internal consistency check. One may verify the coherence of the result: $(M_{\text{PDL}}^*/M_{\text{GR}}^*)^3 = \sigma(40)^{-2} = 1.400726$, confirming that the modified threshold indeed satisfies $\tau_{\text{PDL}}(M_{\text{PDL}}^*) = t_{\text{U}}$ exactly.

4 The observational signature

4.1 Physical picture

PBHs in the mass window $[M_{\text{GR}}^*, M_{\text{PDL}}^*] = [5.10, 5.71] \times 10^{14} \text{ g}$ present the following behaviour under the two frameworks:

- *Standard GR:* $\tau_{\text{GR}}(M) < t_{\text{U}}$ for all $M < M_{\text{GR}}^*$, so these PBHs have fully evaporated and contribute zero flux to the present-day IGRB.

- *PDL*: $\tau_{\text{PDL}}(M) > t_{\text{U}}$ for all $M > M_{\text{PDL}}^*$. PBHs with $M \in [M_{\text{GR}}^*, M_{\text{PDL}}^*]$ have $\tau_{\text{GR}}(M) < t_{\text{U}}$ yet $\tau_{\text{PDL}}(M) > t_{\text{U}}$, so they are still active today and contribute positively to the IGRB.

The PDL prediction therefore implies an *excess of present-day Hawking radiation flux* from the mass window $[M_{\text{GR}}^*, M_{\text{PDL}}^*]$ relative to the GR baseline.

4.2 Spectral localisation

The Hawking temperature of a black hole of mass M is:

$$T_H(M) = \frac{\hbar c^3}{8\pi G M k_B}, \quad (13)$$

which evaluates to:

$$T_H(M_{\text{GR}}^*) \approx 20.7 \text{ MeV}, \quad (14)$$

$$T_H(M_{\text{PDL}}^*) \approx 18.5 \text{ MeV}. \quad (15)$$

The peak photon energy in the Hawking spectrum scales approximately as $E_{\text{peak}} \sim 5 T_H$ [23] for the primary emission channel alone, giving a first-order estimate of $E_{\text{peak}} \approx 93\text{--}104 \text{ MeV}$, already within the Fermi-LAT energy range of 20 MeV to 300 GeV [25].

This estimate considers only the primary/direct emission channel and omits the secondary radiation from hadronisation and decay of unstable particles, the final-state radiation from relativistic electrons and positrons, and the in-flight annihilation of those positrons with the interstellar medium — all of which contribute non-negligibly near the spectral peak [7]. A complete calculation was therefore performed using the public pipeline GAMMAPBHLOTTER [7], which combines all four components and interpolates the primary and secondary spectra from the pre-computed BLACKHAWK [8] tables via a bivariate spline (linear in $\log M$, cubic in $\log E$, fitted to the full grid of 56 pre-rendered masses spanning 5×10^{13} to 10^{19} g). Evaluating the resulting total spectrum at $M = M_{\text{GR}}^* = 5.099 \times 10^{14} \text{ g}$ and at $M = M_{\text{PDL}}^* = 5.705 \times 10^{14} \text{ g}$ gives:

$$E_{\text{peak}}(M_{\text{GR}}^*) \approx 130.1 \text{ MeV}, \quad (16)$$

$$E_{\text{peak}}(M_{\text{PDL}}^*) \approx 117.5 \text{ MeV}, \quad (17)$$

a fractional shift of $\Delta E/E = -9.65\%$ in the position of $E^2 dN_\gamma/dE_\gamma$, both values lying comfortably within the Fermi-LAT range. The shift is close to, but more accurate than, the crude estimate above; the residual difference between 9.65% and the naive expectation $1 - \sigma(40)^{2/3} \approx 10.6\%$ (from $T_H \propto M^{-1}$ alone) reflects the contribution of the secondary and annihilation channels to the position of the total spectral peak. This calculation

supersedes the primary-only estimate used in earlier drafts of this prediction and should be taken as the reference values for any observational test.

4.3 Nature of the signal

The signal is not a shift in the *abundance* of PBHs, which PDL does not predict. It is a shift in the *survival threshold*: the spectral peak of the Hawking contribution to the IGRB is displaced from $E_{\text{peak}}(M_{\text{GR}}^*) \approx 130.1 \text{ MeV}$ to $E_{\text{peak}}(M_{\text{PDL}}^*) \approx 117.5 \text{ MeV}$. In practice, if the initial PBH abundance f_{PBH} is non-zero in the window $[M_{\text{GR}}^*, M_{\text{PDL}}^*]$, one expects:

- An excess of IGRB flux at $E \lesssim 130 \text{ MeV}$ relative to the GR prediction.
- The spectral shape of the excess follows the full multi-component Hawking spectrum [7, 8] evaluated at $T_H \approx 19\text{--}21 \text{ MeV}$.
- The fractional shift in the spectral peak is $\Delta E/E \approx 9.65\%$, comparable to the Fermi-LAT spectral resolution at 100–130 MeV ($\sim 10\text{--}20\%$).

4.4 Additional prediction: entropy suppression

A complementary prediction concerns the integrated entropy-to-energy ratio of PBH-seeded objects at high redshift. The PDL suppression factor at epoch N is $\sigma(N)^2$, so:

$$\frac{S_{\text{PDL}}}{E_{\text{PDL}}} \propto \sigma(N(z))^2 \cdot \left(\frac{S}{E}\right)_{\text{GR}}, \quad (18)$$

predicting a systematic redshift dependence of the entropy-to-energy ratio in AGN populations seeded by PBHs [14]. At z corresponding to $N = 40$ ($z \sim z_{\text{CMB}}$), the suppression factor is $\sigma(40)^2 = 0.7139$, i.e. a 28.6% reduction relative to the local value.

5 How to test this prediction

The PDL prediction can be tested quantitatively using entirely public tools and data.

Step 1 — Compute the Hawking spectra (performed in this revision). The full multi-component photon spectrum $d^2 N_\gamma / (dE dt)$ was computed for monochromatic PBH masses $M = M_{\text{GR}}^* = 5.099 \times 10^{14} \text{ g}$ and $M = M_{\text{PDL}}^* = 5.706 \times 10^{14} \text{ g}$ using GAMMAPBH-PLOTTER [7], which combines BLACKHAWK [8] primary and secondary spectral tables with final-state radiation and in-flight annihilation components. The resulting spectral peaks are reported in Section 4.2 above; the full spectra are available from the authors on request.

Step 2 — Normalise to the IGRB constraints (remaining). Use ISATIS [9], or the constraint pipeline of [12], to compare the computed spectra against the Fermi-LAT IGRB measurement of Ackermann et al. [10] and the constraint compilation of Carr et al. [11]. The relevant quantity is the upper limit on $f_{\text{PBH}}(M)$ in the window $[5.10, 5.71] \times 10^{14}$ g; note that [12] already finds $f_{\text{PBH}} \lesssim 10^{-10}$ near this mass (Section 7).

Step 3 — Identify the discriminating observable. The discriminating quantity is the position of the spectral peak in the 100–150 MeV band of the IGRB. Under GR, this peak occurs at $E_{\text{peak}} \approx 130.1$ MeV; under PDL, it is displaced to $E_{\text{peak}} \approx 117.5$ MeV. A fit to the Fermi-LAT IGRB spectrum in this band with E_{peak} as a free parameter constitutes a direct test of the PDL prediction.

Step 4 — Complementary constraint from the entropy-to-energy ratio. An independent test uses the redshift dependence of the entropy-to-energy ratio in AGN populations. The predicted scaling $S/E \propto \sigma(N(z))^2$ can be confronted with multi-epoch surveys such as those from DESI or Euclid.

6 Relation to the PDL causal chain

The PBH threshold prediction is not an isolated result but a corollary of the causal chain established in the PDL corpus. The complete chain from axioms to the present prediction is:

$$\begin{array}{ccccccc} \text{C1–C4} & \xrightarrow{\text{D16a}} & K_4 & \xrightarrow{\text{D16b, D29}} & \text{quintuplet} & \xrightarrow{\text{D39, D42}} & \kappa_{\text{PDL}} \xrightarrow{\text{D24}} \\ \sigma(N) & \xrightarrow{\text{D31, D36, D42}} & G_{\text{eff}}(N) & \xrightarrow{\text{D27}} & N_{\text{CMB}} = 40 & \xrightarrow{\text{D38}} & M_{\text{PDL}}^* \end{array}$$

Every arrow in this chain has been proved as an unconditional theorem of C1–C4 (see D43v3 [3] for the complete synthesis). The prediction M_{PDL}^* inherits the unconditional character of each link; it contains no free parameter beyond the Standard Model degrees of freedom g^* entering the Hawking spectrum, which are taken from the literature [11].

Position within the PDL corpus. This document constitutes D45 of the PDL corpus. The global mapping of all documents, open problems, and resolved results is maintained in the D-Map [26], currently at version 16.

7 Discussion

What this prediction does and does not claim. The PDL framework predicts the *position* of the PBH survival threshold, not the *abundance* of PBHs in any mass window. The observational signal therefore depends on the initial PBH mass function, which is cosmological model-dependent and not determined by PDL. If f_{PBH} is negligibly small in $[M_{\text{GR}}^*, M_{\text{PDL}}^*]$, the signal will be below any present detection threshold. In that case the prediction cannot be confirmed, but it is also not falsified. Falsification would require demonstrating that the IGRB spectral peak is definitively located at $E_{\text{peak}}^{\text{GR}} \approx 130.1 \text{ MeV}$ rather than at $E_{\text{peak}}^{\text{PDL}} \approx 117.5 \text{ MeV}$, together with a non-negligible f_{PBH} in the relevant window. We note that this latter condition is in tension with a very recent, independent re-analysis of the same IGRB dataset [12], combined with EGRET and COMPTEL data, which constrains $f_{\text{PBH}} \lesssim 10^{-10}$ for PBHs of mass $\sim 10^{14} \text{ g}$ under standard GR evaporation. If this constraint also applies, essentially unchanged, to the PDL-shifted window, the present signal may be too faint to discriminate between the two frameworks with current data, irrespective of which cutoff position is correct; this would not invalidate the prediction, only its near-term testability. We regard this as an open, and honestly reported, limitation of the present proposal rather than a difficulty to be minimised.

Comparison with other modified-gravity scenarios. Several alternative frameworks also predict modifications to the Hawking evaporation rate, including models with extra dimensions [27], memory-burden effects [28], and dark charges [29]. The PDL prediction is distinguished by three features: (i) the shift is derived from a single unconditional combinatorial theorem ($\kappa_{\text{PDL}} \in \mathbb{Q}(\sqrt{5})$), not from a new adjustable parameter; (ii) the same framework simultaneously derives G , α , μ^* , and the Hubble tension resolution; (iii) the shift is positive (heavier threshold) and of a specific magnitude (+11.89% in mass, −9.65% in spectral peak energy) that does not depend on any model tuning.

Sensitivity requirements. The spectral displacement $\Delta E/E \approx 9.65\%$ is at the boundary of current Fermi-LAT sensitivity at $\sim 100\text{--}130 \text{ MeV}$. A dedicated analysis using the full 15+ years of Fermi-LAT data, focusing on the IGRB spectrum in the 80–200 MeV window, would provide the most constraining test, subject to the abundance caveat above. Next-generation MeV telescopes such as AMEGO-X [30] and e-ASTROGAM [31] would improve the sensitivity substantially, potentially resolving the spectral peak position to better than 5%; [12] reaches the same conclusion regarding the need for next-generation instruments in this band.

8 Conclusion

The PDL framework predicts that the primordial black hole survival mass threshold is shifted upward by +11.89% relative to the standard GR value, giving:

$$M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* \approx 5.706 \times 10^{14} \text{ g}. \quad (19)$$

This prediction is an unconditional corollary of the density-dependent effective gravitational coupling $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$, itself proved from the PDL axioms C1–C4 without any free parameter. The resulting observable, computed with the full multi-component Hawking spectrum, is a shift of the spectral peak from $\approx 130.1 \text{ MeV}$ (GR) to $\approx 117.5 \text{ MeV}$ (PDL), a -9.65% displacement in the Hawking contribution to the isotropic gamma-ray background, well within the energy range accessible to Fermi-LAT.

The spectral computation has been performed in this revision using GAMMAPBHPlotTER [7] and BLACKHAWK [8]. What remains is a direct comparison with the Fermi-LAT IGRB data using ISATIS or an equivalent pipeline, subject to the abundance caveat raised in Section 7. This is a standard numerical procedure well within the capacity of existing codes and data. We invite the community to perform this comparison and report the result, whether confirmatory or falsifying.

Falsifiable prediction — Summary

PDL prediction D45: The PBH survival mass threshold satisfies $M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* = 5.706 \times 10^{14} \text{ g}$. The full Hawking flux from PBHs in the window $[5.10, 5.71] \times 10^{14} \text{ g}$ contributes a non-zero signal to the present-day IGRB with spectral peak at $E_{\text{peak}} \approx 117.5\text{--}130.1 \text{ MeV}$ (PDL vs GR, computed with GAMMAPBHPlotTER/BLACKHAWK). This signal is absent in standard GR. It can be tested with BLACKHAWK + GAMMAPBHPlotTER + ISATIS + Fermi-LAT IGRB data, though a recent independent constraint [12] suggests the relevant PBH abundance may be too small for near-term detection.

Acknowledgements

This work is part of the PDL research programme, conducted independently. All corpus documents are open-access on Zenodo under the PDL community (<https://zenodo.org/communities/pdl-framework>) and on GitHub (<https://github.com/laubscher-lab/PDL-framework>).

References

- [1] Cédric Laubscher. The emergence of physical reality within the framework of the projective dynamic logo, 2025.
- [2] Cédric Laubscher. Universal coherence leakage: A structural bridge between G and α in PDL (V3), 2025.
- [3] Cédric Laubscher. The causal chain of physical reality: From four axioms to Newton’s constant via the geometric leakage parameter (PDL framework — D43v3), 2026.
- [4] Cédric Laubscher. Proof of $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ from the trace structure of K_4 and independence of gravitational engagement: Gate 3 strengthened, 2026.
- [5] Cédric Laubscher. Derivation of H3 from axioms C1–C4: The equiparticipation lemma and the closure of causality, 2026.
- [6] Cédric Laubscher. Structural derivation of N_{CMB} from the PDL neutron architecture and parameter-free resolution of the hubble tension, 2026.
- [7] John Carlini and Ilias Cholis. Gammapbhplotter: code and user manual. <https://zenodo.org/records/16944093>, 2025. Zenodo, v1.0.
- [8] Alexandre Arbey and Jérémy Auffinger. BlackHawk: A public code for calculating the Hawking evaporation spectra of any black hole distribution. *The European Physical Journal C*, 79:693, 2019.
- [9] Jérémy Auffinger. Primordial black hole constraints with Hawking radiation — a review. *Progress in Particle and Nuclear Physics*, 131:104040, 2023.
- [10] M. Ackermann et al. Searching for dark matter annihilation from Milky Way dwarf spheroidal galaxies with six years of Fermi large area telescope data. *Physical Review Letters*, 115:231301, 2015. See also: Ackermann et al., ApJ 799, 86 (2015) for the IGRB spectrum measurement used in PBH constraint analyses.
- [11] Bernard Carr, Kazunori Kohri, Yuuiti Sendouda, and Jun’ichi Yokoyama. Constraints on primordial black holes. *Reports on Progress in Physics*, 84(11):116902, 2021.
- [12] Ilias Cholis, Iason Krommydas, and John Carlini. Limits on primordial black holes from the extragalactic gamma-ray background; current status and future projections. *arXiv preprint*, 2026. arXiv:2606.10013.
- [13] Cédric Laubscher. Derivation of $\kappa = R_{\text{surf}}/R_{\text{tot}}$ from the PDL axioms: the indifference lemma and partial resolution of OP1 (D36), 2026.

- [14] Cédric Laubscher. Bekenstein–Hawking entropy from the PDL effective gravitational coupling: the 4π identity, surface counting, and primordial black hole predictions, 2026.
- [15] Cédric Laubscher. Introduction to the projective dynamic logo (PDL), 2025.
- [16] Cédric Laubscher. Minimal stationary closures in the PDL framework: Necessity of the (4,6) block, 2025.
- [17] Cédric Laubscher. On the combinatorial selection and local uniqueness of the PDL proton architecture, 2025.
- [18] Cédric Laubscher. Axiomatic derivation of the leading valence engagement amplitude in the PDL framework: Gate 1 resolved, 2026.
- [19] Cédric Laubscher. Closure-density dependence of the effective gravitational coupling and the structural origin of the hubble tension, 2025.
- [20] Cédric Laubscher. Proof of $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ in PDL: Gate 3 resolved (preliminary version), 2026.
- [21] Cédric Laubscher. Quantitative Einstein equation from $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ in the PDL framework, 2026.
- [22] S. W. Hawking. Particle creation by black holes. *Communications in Mathematical Physics*, 43:199–220, 1975.
- [23] J. H. MacGibbon and B. J. Carr. Cosmic rays from primordial black holes. *The Astrophysical Journal*, 371:447–469, 1991.
- [24] N. Aghanim et al. Planck 2018 results. VI. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020.
- [25] W. B. Atwood et al. The large area telescope on the Fermi gamma-ray space telescope mission. *The Astrophysical Journal*, 697:1071–1102, 2009.
- [26] Cédric Laubscher. PDL global mapping of structures, results, and open problems (version 16), 2026.
- [27] Luis A. Anchordoqui, Ignatios Antoniadis, and Dieter Lüst. Primordial black holes from conformally coupled scalar fields and Reissner–Nordström black holes in brane world scenarios. *Physical Letters B*, 824:136816, 2022.
- [28] V. Thoss, A. Burkert, and K. Kohri. Breakdown of Hawking evaporation opens new mass window for primordial black holes as dark matter candidate. *Monthly Notices of the Royal Astronomical Society*, 532:451–466, 2024.

- [29] Mrunal Korwar and Stefano Profumo. Updated constraints on primordial black hole evaporation. *Journal of Cosmology and Astroparticle Physics*, 2023:054, 2023.
- [30] Julie McEnery et al. All-sky Medium Energy Gamma-ray Observatory: exploring the obscured universe. *Bulletin of the American Astronomical Society*, 51:7, 2019.
- [31] A. De Angelis et al. The e-ASTROGAM mission (exploring the extreme universe with gamma rays in the MeV–GeV range). *Experimental Astronomy*, 44:25–82, 2017.